Tighter financing goes with lower ESG scores in Taiwan’s food industry

Financial constraints (SA index) and ESG performance (TEJ TESG score): 24 TWSE-listed food firms, 219 firm-years, 2016–2025

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Ming Chuan University, Taiwan · November 2026

Higher financial constraints go with lower ESG scores, supporting H1, but only as a between-firm pattern tied to firm size

−4.805

TESG points per one-unit rise in the SA index

p = 0.0498 (t(23) critical values)

219 firm-years, 24 firms

Random effects, year effects, SE clustered by firm

ROADMAP

Introduction → Methods → Results → Discussion

WHAT SUPPORTS H1

- Negative and significant at 5%: −4.805 (p = 0.0498). The one-year lagged index is also negative: −3.745 (p = 0.055)
- The sign is negative in all four robustness checks
- Concentrated in the social pillar: −8.274 (p = 0.002)

WHAT LIMITS IT

- A between-firm pattern: the fixed-effects estimate is +2.245, because SA barely moves within a firm
- Tied to firm size: adding SIZE cuts the coefficient to −2.519 (p = 0.260)
- Only 24 firms, and p = 0.0498 sits right at the 5% line

Source: TEJ; own calculations. Table 4 and Table 5 of the report.

ESG research asks whether ESG pays; whether a firm’s finances shape its ESG is much less studied

What we know

≈ 90%

of about 2,200 studies report a nonnegative link between ESG and financial performance (Friede et al., 2015, p. 210). Yet hypotheses and results still conflict (Gillan et al., 2021, p. 1).

What is missing

Reverse link

Little work asks whether a firm’s financial position shapes how much it invests in ESG. Taiwan evidence runs the other way: Chang et al. (2025) regress constraint indices on lagged ESG scores, TWSE and OTC firms, 2015–2021.

Why food firms

Since 2014

TWSE-listed food companies must publish CSR reports and have them assured (Tseng, 2022, p. 487). TEJ’s TESG score, launched in 2022, rates all listed Taiwanese firms.

RESEARCH QUESTION

Do financial constraints affect the ESG performance of TWSE-listed food firms in 2016–2025, and in which direction?

Sources: Friede et al. (2015); Gillan et al. (2021); Chang et al. (2025); Tseng (2022); TEJ (2022). Introduction of the report.

Two theories predict opposite signs for the same coefficient, so the data must decide

H1 (−) ESG is a discretionary cost

Higher financial constraints lead to lower ESG scores: firms cut ESG spending to conserve cash.

Fazzari et al. (1988): internal and external funds are not perfect substitutes, so spending depends on financial condition. Hong et al. (2012): less constrained firms have higher corporate goodness scores.

H2 (+) ESG opens access to finance

Higher financial constraints lead to higher ESG scores: firms invest in ESG to attract ESG-conscious capital and lower the cost of funds.

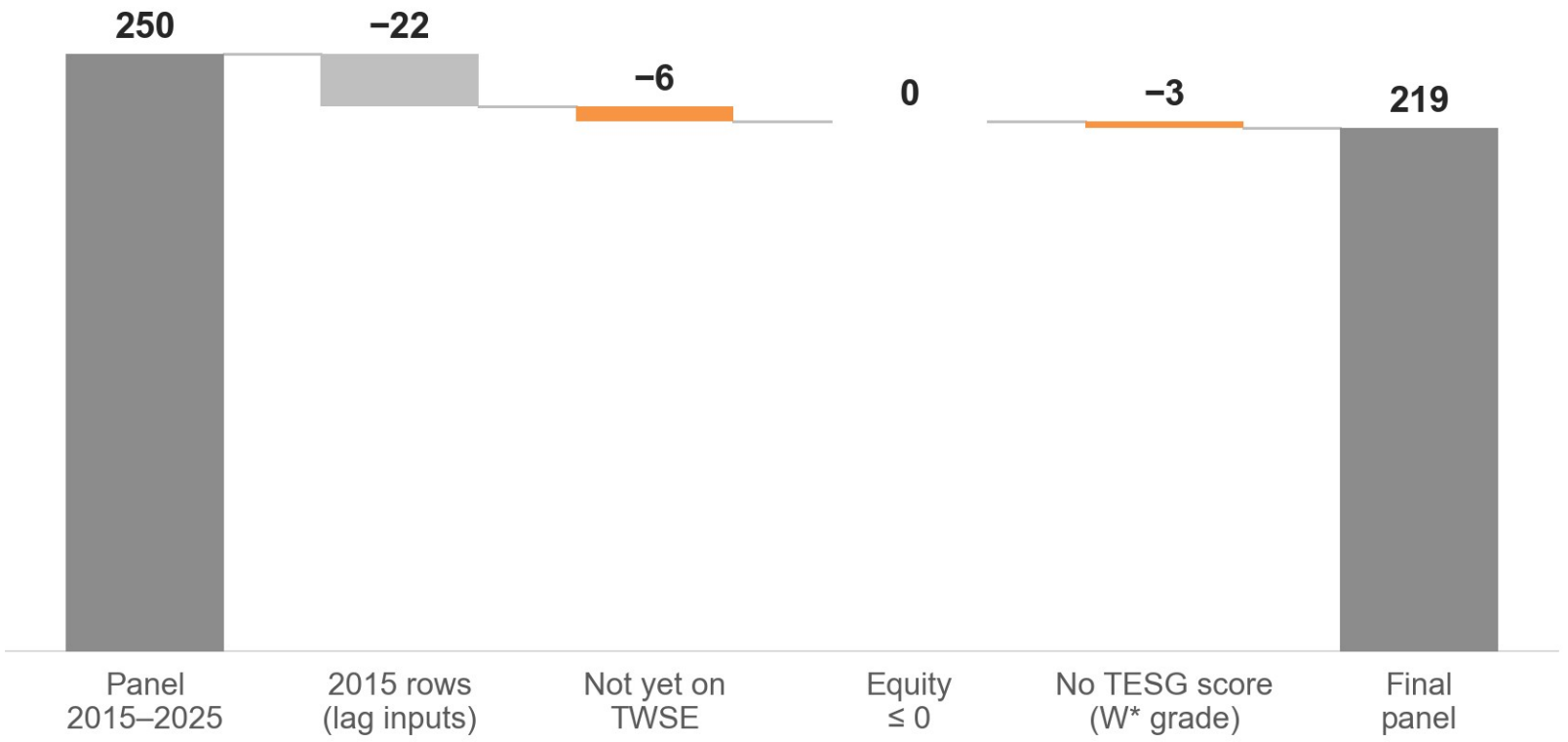
Cheng et al. (2014): firms with better CSR performance face significantly lower capital constraints.

The test: the sign of β_1 on the SA index. $\beta_1 < 0$ supports H1; $\beta_1 > 0$ supports H2.

Contribution: industry-level evidence on the direction from financial constraints to ESG, for a sector with mandatory, assured sustainability reporting.

Sources: Fazzari et al. (1988); Hong et al. (2012); Cheng et al. (2014). See the report's references.

The panel keeps 219 of 250 rows: 22 are 2015 lag inputs and only 9 fail the sample rules



39 → 24 firms

TEJ returned 39 food entries for TWSE. Five depositary receipts and ten firms delisted before 2016 are removed. None left during 2016–2025.

22 rows set aside

2015 data are used only as lagged inputs (GROWTH and lagged SA in 2016). They never enter a regression.

3 firms, no 2025 score

1218, 1219 and 1225 carry the W* warning grade (a prosecutor's indictment). They are not a random group.

Source: TEJ; own calculations. Table 1 and Section 3.1 of the report; data_clean/sample_attrition.csv.

The outcome is the TESG score and constraints are measured by the SA index, built from size and age

Outcome: TESG score

- TEJ composite score, 0–100: fully quantitative, three pillars, sixteen themes
- E, S and G pillar scores serve as alternative outcomes
- December rating, merged to the year of the sustainability report it uses
- Numeric score used throughout; winsorized at 1% and 99%

Constraint: SA index

SA = −0.737 × Size + 0.043 × Size² − 0.040 × Age

- Size: ln of total assets in US$ million, at each year's average USD/NTD rate
- Age: observation year minus founding year
- Both winsorized at 1%/99%; SA is negative in every observation (−6.002 to −2.540); higher means tighter constraints

Controls

- ROA: net income after tax / total assets
- LEV: total liabilities / total assets
- TQ: (market capitalisation + total liabilities) / total assets
- GROWTH: revenue change / previous-year revenue
- CASH: cash and equivalents / total assets
- SIZE (ln total assets) enters only Model 2, because it is part of SA

Source: TEJ; own calculations. Sections 3.3.1 to 3.3.3 and Appendix Table A2 of the report.

The model follows the Hausman rule, uses firm-clustered errors and is stress-tested four ways

$$\mathbf{TESG_{it}} = \alpha_i + \beta_1 SA_{it} + \beta_2 ROA + \beta_3 LEV + \beta_4 TQ + \beta_5 GROWTH + \beta_6 CASH + \gamma_t + \varepsilon_{it}$$

α: firm effect; γ: year effects. SIZE is left out of the main model because ln(total assets) is already a component of SA.

1 Choose the estimator

Hausman test, fixed versus random effects, on the six slope coefficients: $p < 0.05$ → fixed effects; $p \geq 0.05$ → random effects is acceptable.

Mundlak test as a check: firm means of the regressors added to the random-effects model.

2 Estimate the main model

Firm effects, year effects (dummies), standard errors clustered by firm (CR1).

With 24 firms, stars use t(23) critical values (Cameron and Miller, 2015, p. 340).

$\beta_1 < 0$ supports H1; $\beta_1 > 0$ supports H2.

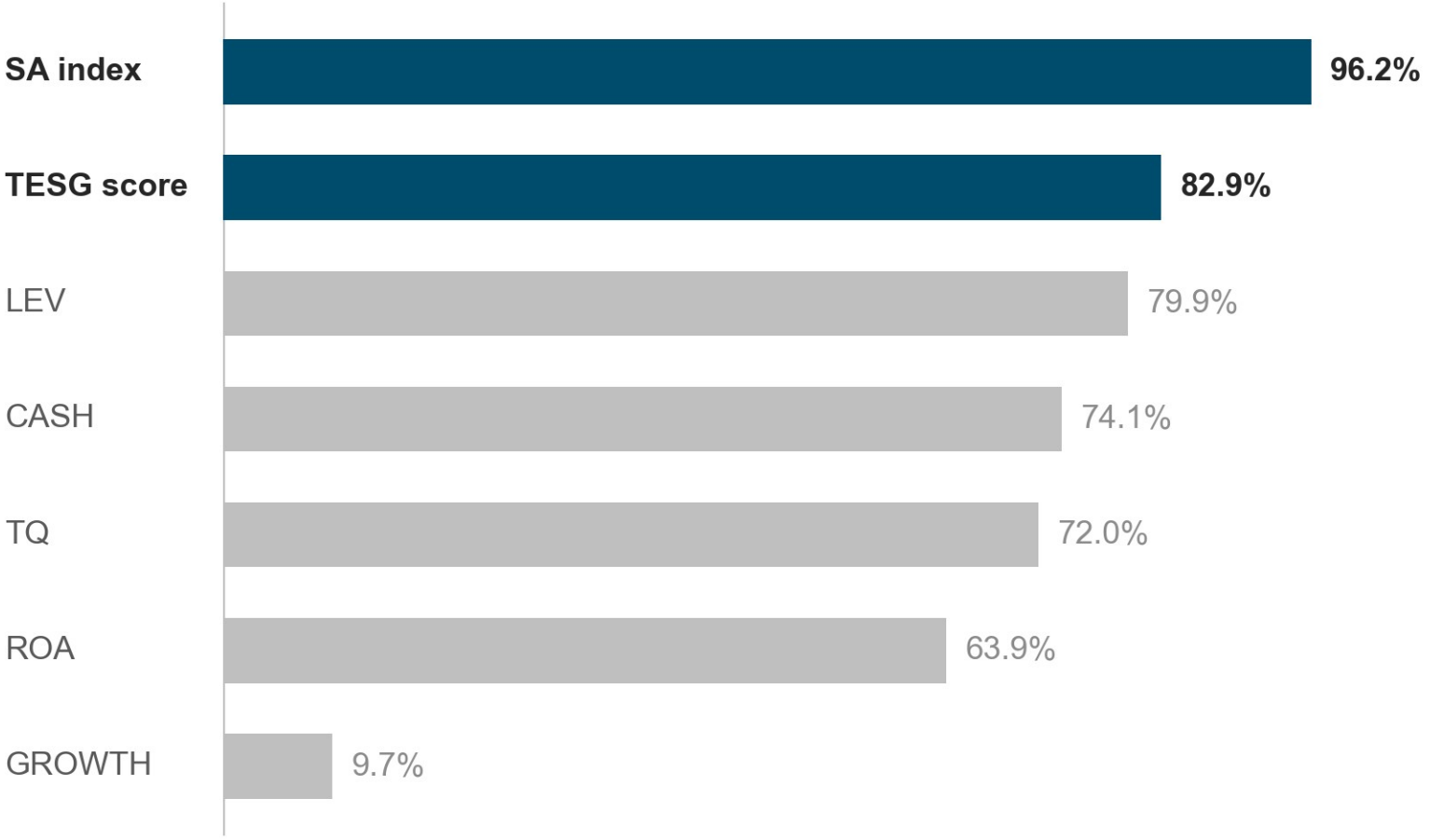
3 Stress-test it

Four checks with the same estimator and inference:

- SA dummy: 1 if above the sample median
- E, S and G regressions, one pillar at a time
- SA index lagged one year
- SIZE added as a control (Model 2), with VIFs

Source: TEJ; own calculations. Sections 3.4 and 3.5 of the report.

Most variation lies between firms: 83% of TESG and 96% of SA variance



0.32%

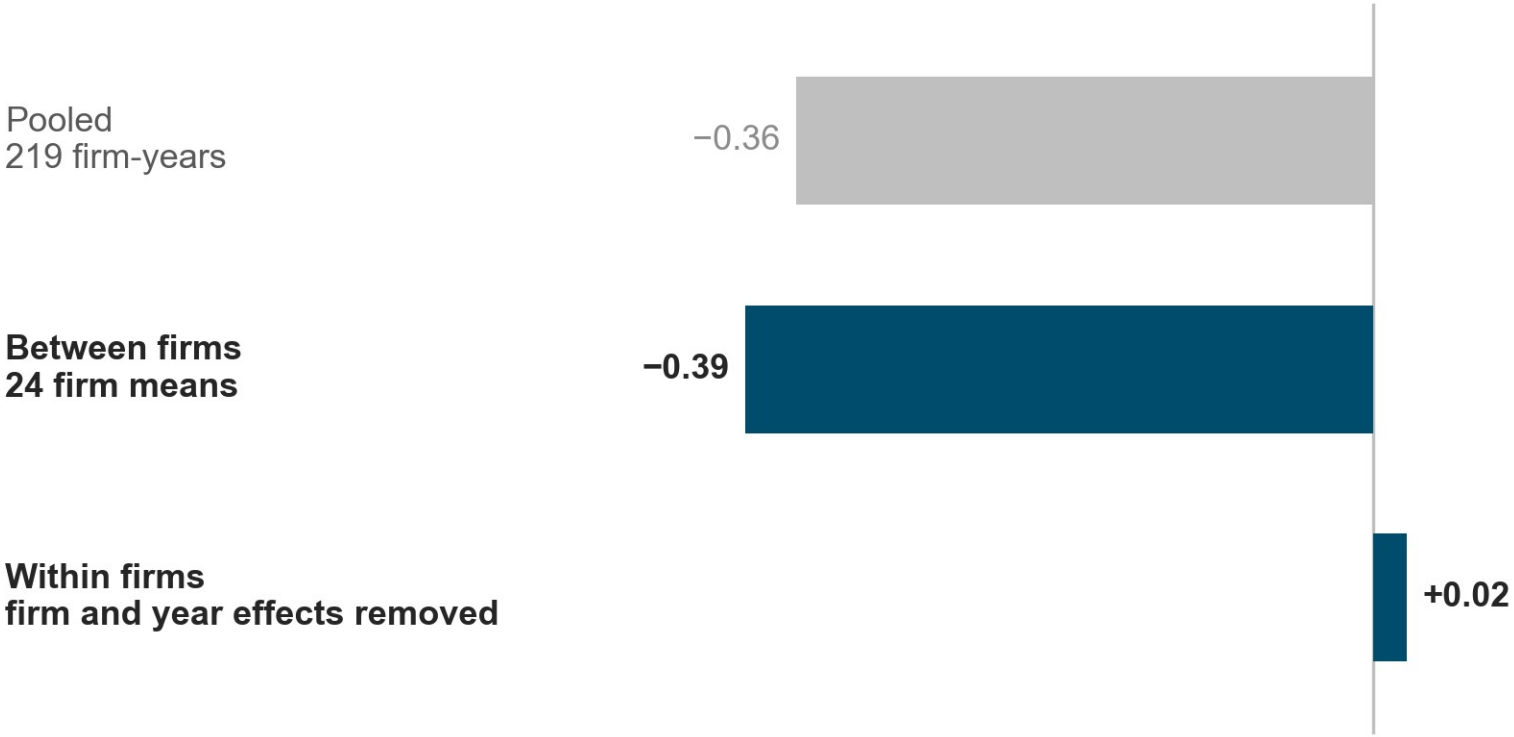
of the variance in SA is left after removing firm and year means: fixed effects have almost nothing to work with.

Age + year

Age is calendar year minus founding year, so within a firm it rises one for one with time and the SA index barely moves.

Source: TEJ; own calculations. Share of each variable's variance between firms; winsorized variables; Section 4.1 of the report.

The link lives between firms: r = −0.39 across firm means, +0.02 once firm and year effects are removed



r = −0.39

across the 24 firm means: firms with tighter constraints score lower.

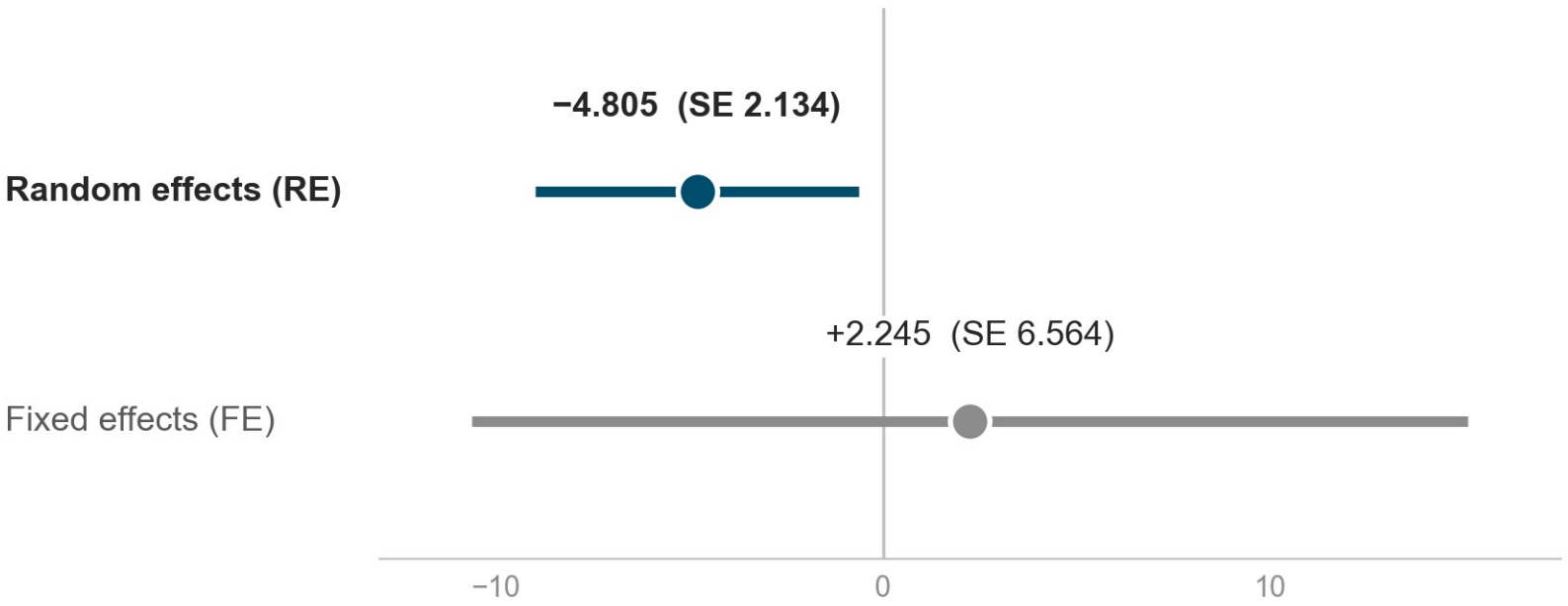
r = +0.02

once firm and year effects are removed: no link inside a firm over time.

So any estimate will say how firms with different constraint levels differ in ESG, not how one firm's ESG changes when its own constraints change. Hadlock and Pierce (2010, p. 1931) expect this: the index captures cross-sectional rather than time variation.

Source: TEJ; own calculations. Figure 1 of the report; outputs/tables/eda_corr_decomposition.csv.

Tests do not reject random effects, and the fixed-effects SA estimate is three times less precise



Whiskers: ±1.96 conventional standard errors, as used in the Hausman test

Decision rule: p ≥ 0.05, so random effects is acceptable and is used for the main regression and every robustness check.

p = 0.816

Hausman $\chi^2(6) = 2.94$: no evidence that firm effects are correlated with the regressors. Mundlak agrees: $\chi^2(6) = 3.25$, p = 0.776.

SE 6.564 vs 2.134

Fixed effects uses only the tiny within-firm variation in SA. Random effects also uses differences between firms, where most of it lies.

Source: TEJ; own calculations. Appendix Table A3 of the report. Whiskers use conventional standard errors; Table 4 uses firm-clustered ones.

A one-unit rise in the SA index goes with a 4.8-point lower TESG score (p = 0.0498)

−4.805

TESG points per one-unit rise in the SA index (p = 0.0498)

≈ −3.82 points

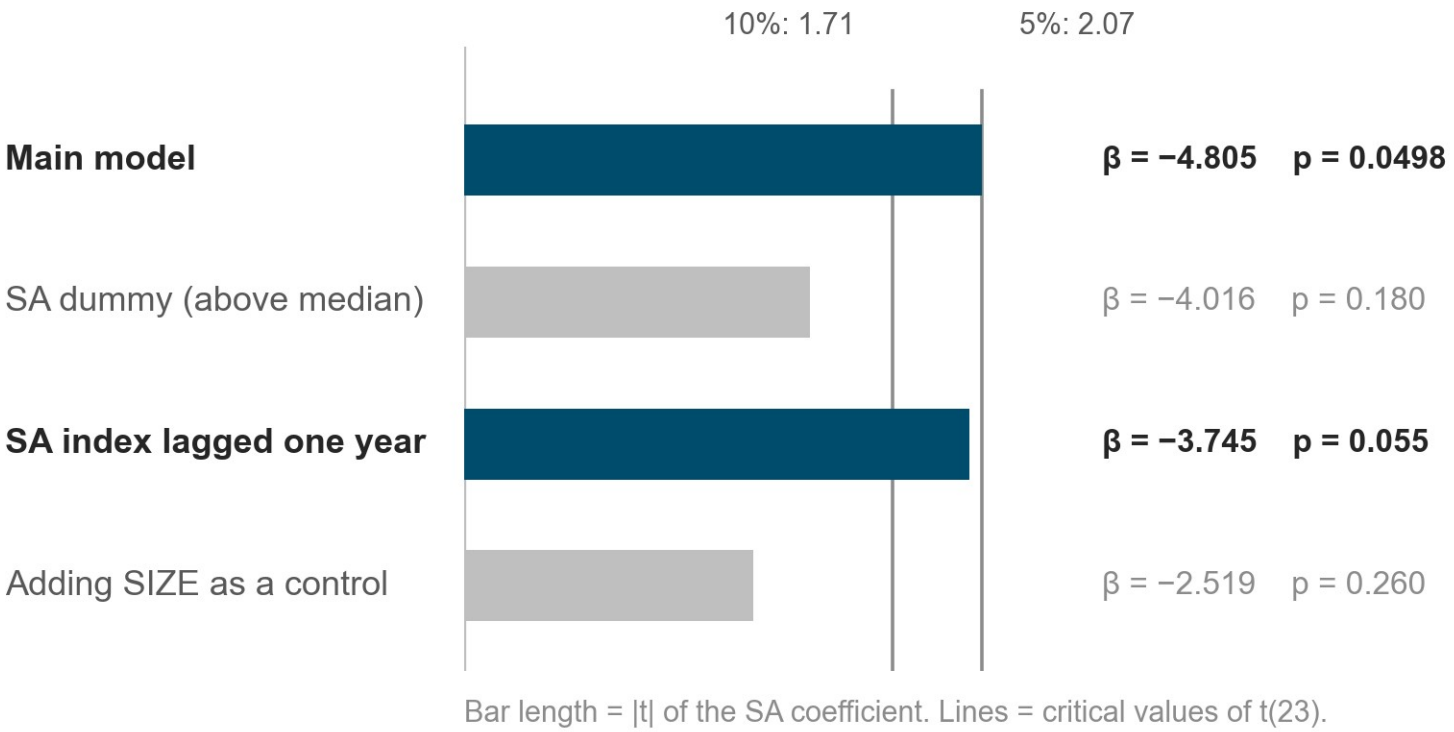
for one between-firm standard deviation of SA (0.796): about 0.45 of the between-firm standard deviation of TESG (8.47).

No control variable is significant at the 10% level.

	Coefficient	SE
SA Index	**−4.805****	**(2.320)**
ROA	3.995	(10.693)
LEV	−7.327	(5.972)
TQ	0.851	(1.807)
GROWTH	−3.301	(2.305)
CASH	8.712	(6.369)
Firm effects	Random	
Year effects	Yes	
Observations	219	
Firms	24	
Overall R^2	0.180	

Source: TEJ; own calculations. Table 4 of the report. Stars: *** p<0.01, ** p<0.05, * p<0.1, t(23) critical values, SE clustered by firm.

The sign holds in all four checks; only the main model (5%) and the lagged index (10%) are significant



Bar length = |t| of the SA coefficient. Lines = critical values of t(23).

SIZE

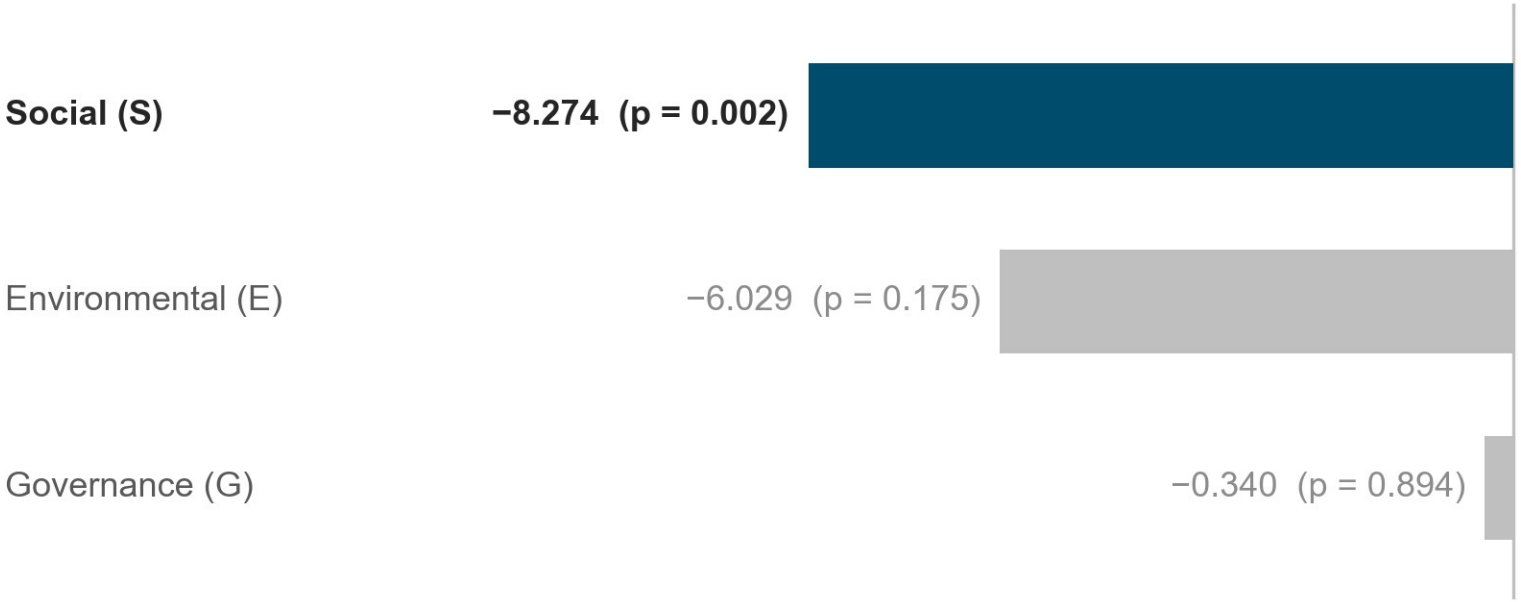
Adding it cuts SA to −2.519 (p = 0.260); SIZE is +2.222 (p = 0.231). SA and SIZE correlate at −0.59.

Lag

Constraints measured a year earlier also go with lower ESG (−3.745, p = 0.055), which eases reverse-causality concerns.

Source: TEJ; own calculations. Table 5 of the report. Critical values of t(23): 1.71 at 10%, 2.07 at 5%.

The association sits in the social pillar (−8.27, p = 0.002); E and G are not significant



Leverage too

LEV on the social score: −13.439 (p = 0.021). More indebted firms also score lower on social performance.

Consistent

Cheng et al. (2014) and Chang et al. (2025) also find the social dimension among the most consistent links with constraints.

Read as suggestive: four scores are tested, and the pillar models use the main specification without a SIZE control.

Source: TEJ; own calculations. Table 6 of the report.

The result fits the discretionary-cost view and repeats the size sensitivity seen in earlier Taiwan work

Fits H1, the discretionary-cost view

Less constrained firms spend more on corporate goodness (Hong et al., 2012, p. 2), and higher ESG performance is most prevalent in larger firms with more resources (Gillan et al., 2021, p. 10).

Describes differences between firms

The random-effects estimate draws mainly on how firms with different constraint levels differ, not on how one firm’s ESG moves when its constraints change, as Hadlock and Pierce (2010, p. 1931) expect of the SA index.

Repeats the size sensitivity of earlier Taiwan work

Chang et al. (2025, p. 5) report a similar sensitivity of the SA index to firm size; here the coefficient falls from −4.805 to −2.519 once SIZE is added.

Points to the social dimension

In line with Cheng et al. (2014, p. 1), whose results are driven by the social and environmental dimensions, and with Chang et al. (2025, p. 6).

Sources: Hong et al. (2012); Gillan et al. (2021); Hadlock and Pierce (2010); Chang et al. (2025); Cheng et al. (2014). Sections 4.3 and 4.4 of the report.

Read it as an association between firms, not as a causal effect of constraints on ESG

Size, or constraints?

SA is built from size and age. With SIZE added the coefficient falls to −2.519 (p = 0.260); the two cannot be separated in this sample (r = −0.59).

Between firms, not within

The fixed-effects estimate is +2.245 (SE 6.564). Random effects assumes firm traits are uncorrelated with the regressors, a strong assumption when SA is close to a function of size.

Few clusters

24 firms, so inference rests on few clusters. We use t(23) critical values, and p = 0.0498 sits right at the 5% line.

Data and causality

TESG for 2016–2020 was computed retrospectively in 2022; three W*-graded firms have no 2025 score; the firm list is TEJ’s current table. Estimates are associations: reverse causality is not resolved.

Source: TEJ; own calculations. Section 5 of the report.

Constrained firms score lower on ESG, so low scores may partly reflect tight finance

ANSWER TO THE RESEARCH QUESTION

Yes: firms with tighter financial constraints have lower ESG scores, supporting H1.

β = −4.805, p = 0.0498. ESG behaves as a discretionary cost rather than as a signal for access to finance (H2); the pattern is between firms and cannot be separated from firm size.

IMPLICATIONS

- Within the food industry, ESG performance is weaker where financing is tighter, particularly on the social dimension
- Investors and lenders that use ESG scores should know that low scores may partly reflect financial constraints, not only management choices
- Policy that eases access to finance for smaller listed firms may support their sustainability performance

NEXT STEPS

- Constraint measures less tied to size, such as the KZ and WW indices (Cheng et al., 2014; Chang et al., 2025)
- A cross-industry sample of Taiwanese firms, to add clusters

Source: TEJ; own calculations. Section 5 of the report.

A. TESG averages 60.5 (SD 9.2) and the SA index is negative in all 219 observations

	N	Mean	SD	Min	Median	Max
Panel A: Model 1 variables						
TESG score	219	60.547	9.227	36.239	61.850	76.511
SA Index	219	−4.765	0.735	−6.002	−4.761	−2.540
ROA	219	0.056	0.049	−0.071	0.051	0.217
LEV	219	0.401	0.167	0.118	0.384	0.734
TQ	219	1.452	0.658	0.759	1.246	3.587
GROWTH	219	0.037	0.135	−0.333	0.025	0.560
CASH	219	0.115	0.089	0.002	0.099	0.413
Panel B: Other variables						
E score	219	61.061	15.049	30.806	62.330	85.325
S score	219	64.343	10.825	38.857	66.410	85.098
G score	219	56.192	9.285	30.927	56.290	72.616
SIZE (ln total assets, NTD thousand)	219	16.250	1.310	13.623	16.059	20.283
Size (ln total assets, US$ million)	219	5.924	1.311	3.269	5.757	9.943
Age (years)	219	49.553	15.181	7.180	51.000	73.820

Source: TEJ; own calculations. Table 2 of the report; N = 219 firm-years, 24 TWSE-listed food firms, 2016–2025.

B. No pair of variables exceeds |r| = 0.7; SA and SIZE correlate at −0.59

	TESG	SA	ROA	LEV	TQ	GROWTH	CASH	SIZE†
TESG	1							
SA	−0.36***	1						
ROA	0.19***	0.23***	1					
LEV	0.11*	−0.37***	−0.27***	1				
TQ	0.05	0.35***	**0.61*** **	0.02	1			
GROWTH	−0.06	−0.01	0.15**	−0.05	0.20***	1		
CASH	0.15**	0.21***	0.48***	−0.17***	0.32***	0.04	1	
SIZE†	**0.58*** **	**−0.59*** **	0.11*	0.34***	−0.08	0.05	0.01	1

−0.59

SA and SIZE (ln total assets) are strongly negatively correlated: SIZE is a component of SA, so it is tested only in Model 2.

VIF ≤ 2.37

Largest VIF is 2.13 in Model 1 and 2.37 in Model 2 (ROA): multicollinearity is not a concern.

Source: TEJ; own calculations. Table 3 of the report. *** p<0.01, ** p<0.05, * p<0.1. † SIZE = ln(total assets), Model 2 only.

C. Hausman (p = 0.816) and Mundlak (p = 0.776) tests do not reject random effects

	FE	RE	Difference	SE (FE)	SE (RE)
SA Index	**2.245**	**−4.805**	**7.049**	**6.564**	**2.134**
ROA	1.048	3.995	−2.947	11.231	10.922
LEV	−8.851	−7.327	−1.524	4.186	3.973
TQ	0.793	0.851	−0.058	0.887	0.863
GROWTH	−3.599	−3.301	−0.298	2.376	2.303
CASH	7.329	8.713	−1.384	6.464	6.170

Test	Statistic	df	p-value
Hausman (classic)	2.94	6	0.816
Mundlak	3.25	6	0.776

Both models include year dummies and use conventional covariance matrices, as the Hausman test requires, so the standard errors differ from the clustered ones in Table 4. The covariance difference is positive definite, so the statistic is valid.

Source: TEJ; own calculations. Appendix Table A3 of the report.

D. Every variable is built from TEJ fields as defined here

Variable	Definition
TESG	TEJ TESG composite score (0–100), December rating merged to the year of the sustainability report it uses; winsorized 1%/99%
E, S, G	TEJ environmental, social and governance pillar scores (0–100), same timing as TESG; winsorized 1%/99%
Size	ln of total assets in US$ million, converted at each year’s average USD/NTD rate; winsorized 1%/99%
Age	Observation year minus founding year; winsorized 1%/99%
SA Index	$-0.737 \times \text{Size} + 0.043 \times \text{Size}^2 - 0.040 \times \text{Age}$, from winsorized Size and Age (Hadlock & Pierce, 2010)
SA dummy	1 if SA is above the pooled sample median, 0 otherwise
ROA	Consolidated net income after tax / total assets
LEV	Total liabilities / total assets
TQ	(Market capitalisation + total liabilities) / total assets
GROWTH	(Revenue t − Revenue t−1) / Revenue t−1, within firm
CASH	Cash and cash equivalents / total assets
SIZE	ln of total assets in NTD thousand; Model 2 only; winsorized 1%/99%

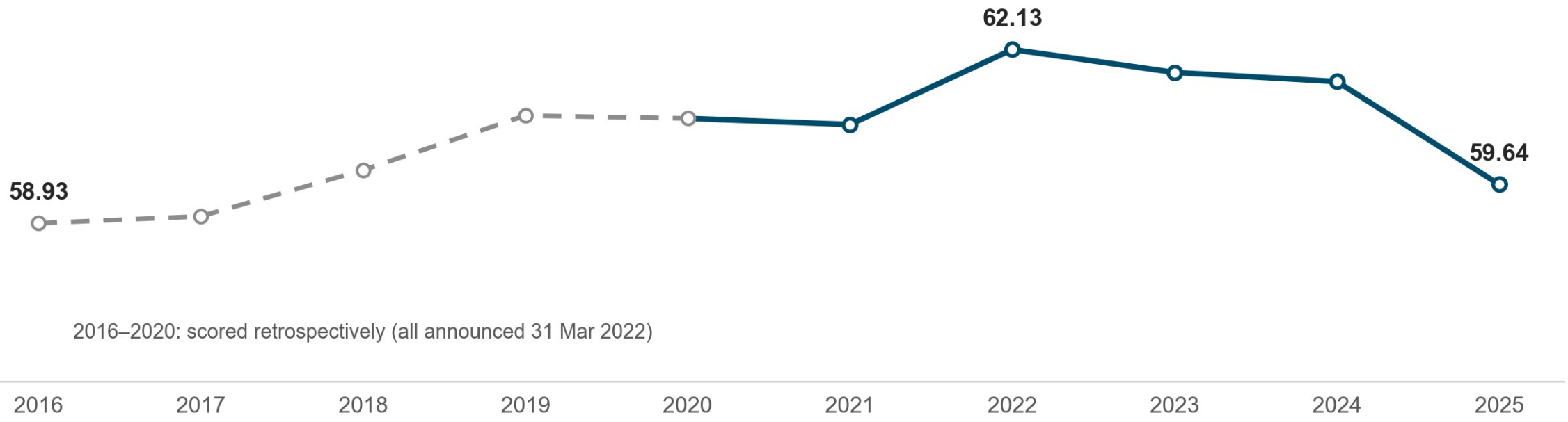
Source: TEJ; own calculations. Appendix Table A2 of the report (TEJ field names are listed there).

E. The 24 sample firms: 19 have ten observations, 3 have nine and 2 enter only in 2025

Code	Firm	Founded	TWSE listing	Firm-years
1201	WEI CHUAN	1953	1962	10
1203	Ve Wong	1959	1964	10
1210	GREATWALL	1960	1978	10
1213	Oceanic Bev.	1965	1981	10
1215	CPE	1977	1987	10
1216	Uni-President	1967	1987	10
1217	AGV	1971	1989	10
1218	Taisun	1950	1989	9
1219	FWUSOW	1955	1990	9
1220	Tairoun	1969	1991	10
1225	FOPCO	1986	1993	9
1227	SFC	1986	1994	10
1229	LHIHC	1955	1976	10
1231	Lian Hwa Foods	1970	1995	10
1232	TTET	1982	1996	10
1233	Ten Ren	1975	1999	10
1234	HEY SONG	1969	1999	10
1235	ST	1972	2000	10
1236	Hunya	1976	2001	10
1256	Sunjuice	2010	2016	10
1702	Namchow	1950	1973	10
1737	Taiyen	1995	2003	10
7780	Daiken	2018	2025	1
7791	Lucky Royal	1986	2025	1

Source: TEJ; own calculations. Appendix Table A1 of the report. Founded = year of establishment (used for Age).

F. Average TESG rose to 62.1 by 2022 and fell back to 59.6 in 2025; scores before 2021 are retrospective



Source: TEJ; own calculations. Section 4.1 of the report; outputs/tables/eda_by_year.csv, eda_tesg_announcement.csv.